



DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2024-25
PRE BOARD EXAMINATION

CLASS: X
SUBJECT: GEOGRAPHY (SET A)

FULL MARKS: 80
TIME: 2 HOURS

*Answers to this Paper must be written on the paper provided separately.
You will not be allowed to write during the first 15 minutes.
This time is to be spent in reading the question paper.*

*The time given at the head of this Paper is the time allowed for writing the answers.
Part I is compulsory. All questions from Part I are to be attempted.
A total of five questions are to be attempted from Part II.
The marks intended for questions are given in brackets [].
This paper consists of seven printed pages.*

PART I (30 marks)
(Attempt all questions from this Part.)

Question 1

Study the map extract G43S7 of the Survey of India and answer the following questions:

- (i) (a) Give the six figure grid reference of Δ 233 and surveyed tree near spot height 189. [2]
(b) What does the white patch (3892) in the main river indicate?
- (ii) Nita goes on a bicycle from 86 northing to 94 northing in a straight line. If she moves at a speed of 10 kilometres per hour, how much time will she take to reach the destination? [2]
- (iii) (a) What is the compass direction of Sangla (3786) from Vaghol (2988)? [2]
(b) Calculate the area in sq. Kilometre within the easting 30 to 37 and Northing 89 to 93.
- (iv) What do the following mean? [2]
(a) 10r in grid square 3285.
(b) 106 in grid square 3187
- (v) Calculate the indirect distance along the metalled road between Jorapura(2991) and Ramsida(3190)in km. [2]

Question 2

On the outline map of India provided:

- (i) Mark with arrows and name the wind bringing rainfall to Kochi in monsoon. [1]
- (ii) Mark and name the electronic capital of India. [1]
- (iii) Mark and name the landlocked port city of East Coast. [1]
- (iv) Mark and name the river flowing through the rift valley. [1]
- (v) Mark and name the peak located on the north-western part of Sikkim. [1]
- (vi) Shade and name the state of highest population density below Tropic of Cancer. [1]
- (vii) Mark and name the offshore oilfield in India. [1]
- (viii) Shade and label Wular Lake. [1]
- (ix) Mark and label the most important latitude passing through the country. [1]
- (x) Mark and name the strait between India and Sri Lanka. [1]

Question 3

Choose the correct answers to the questions from the given options.

[10]

(Do not copy the question, write the correct answers only).

(i) The shaded portion of the given outline map of India represents



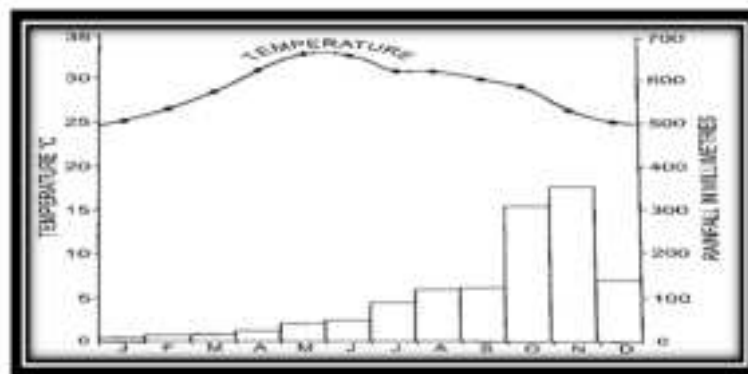
(a) Areas affected by drought

(b) Areas under cotton cultivation

(c) Areas under wheat cultivation

(d) Sparsely populated areas

(ii) This climatic graph is showing the pattern of rainfall and temperature of a station which receives maximum rainfall in the month of November. What could be a possible reason for this pattern?



(a) Coastal winds from the ocean

(b) High altitude climate

(c) Influence of polar air masses

(d) Influence of monsoon winds

(iii) The Western Ghats Region (WGR) is a globally significant biodiversity hotspot. Its importance was recognised by UNESCO in 2012. The region is an important agricultural area, with a significant portion of the land being used for the cultivation of tea, coffee, rubber, palm and other crops, as well as for livestock grazing and roads. The portion of WRG in Gujarat had registered soil erosion of 34,4,69,1,71.1 and 75.3 tonnes per hectare per year for years 1990, 2000, 2010 and 2020 respectively. What type of erosion is more prevalent in WGR?

(a) Wind erosion

(b) Unscientific method of agricultural practice

(c) Sheet erosion due to heavy downpour

(d) Riverbank erosion

(iv) Match the following:

LIST I	LIST II
A. Copper	1. Noamundi
B. Coal	2. Sundergarh
C. Manganese	3. Balaghat district
D. Iron Ore	4. Jharia

CODES

	A	B	C	D
(a)	3	4	2	1
(b)	1	2	3	4

	A	B	C	D
(c)	3	4	1	2
(d)	4	3	2	1

(v) Which Steel plant amongst the following was set up with the collaboration with USSR?

- (a) Bhilai
- (b) Tata Steel
- (c) Vishakhapatnam
- (d) Rourkela

(vi) Which irrigation techniques are being promoted in India to reduce water dependency?

- (a) Rainwater harvesting and contour ploughing
- (b) Drip and Sprinkler irrigation
- (c) Canal System
- (d) Cloud seeding and artificial rainfall

(vii) The Green Revolution in the 1960s and 1970s marked a turning point in Indian agriculture, dramatically increasing the yield of crops through the use of high-yielding varieties (HYVs), chemical fertilizers, and pesticides. Primarily benefiting states with established irrigation infrastructure are Punjab and Haryana.

(Edited;Source: Editorial is based on the Article Man of Science and Humanity;
Published in The Indian Express on 29/09/2023)

Which two crops had the highest productivity gains during India's Green Revolution?

- (a) Sugarcane and maize
- (b) Wheat and pulses
- (c) Cotton and millets
- (d) Wheat and rice

(viii) A major advantage of incineration over open dumping is –

- (a) Reduces waste volume significantly
- (b) Increases water retention in soil
- (c) Produces no air emission
- (d) More cost – effective for rural areas

(ix) The Indian government is accelerating expressway construction to boost connectivity across key regions. One prominent project is the Delhi-Mumbai Expressway, stretching over 1,386 km to connect the national capital with the financial hub, Mumbai.

What is the primary purpose of the Delhi – Mumbai Expressway?

- (a) To connect tourist spots across North India
- (b) To reduce travel time between Delhi and Mumbai
- (c) To provide direct route to religious sites
- (d) To promote intercity bus services

(x) Assertion: Mysore is the centre of silk industry.

Reasoning: Mysuru's climate is suitable for mulberry cultivation.

(a) Both Assertion and Reason are true, but Reason is NOT the correct explanation of the Assertion.

(b) Both Assertion and Reason are true, but Reason is the correct explanation of the Assertion.

(c) Assertion is true, but Reason is false.

(d) Both Assertion and Reason are incorrect.

PART II (50 marks)

(Attempt any five questions from this Part)

Question 4

(i) The IMD recently highlighted the increasing frequency of cyclonic storms in the Bay of Bengal, attributing the trend to rising sea surface temperatures. Cyclones have been hitting coastal regions with greater intensity, causing widespread damage to agriculture, infrastructure, and livelihoods. [2]

— Hindustan Times, 10.09.2023

(a) Explain how rising sea surface temperatures contribute to the formation of cyclones.

(b) Why are tropical cyclones more common in Bay of Bengal?

(ii) How does El-Nino affect the climate of India? From where does the winter showers in Western Uttar Pradesh originate? [2]

(iii) Give a reason for each of the following: [3]

(a) Kerala has the longest rainy season.

(b) Rishita frequently travels between Kochi and Jaipur and notices that the number of dry days is higher in Jaipur than in Kochi.

(c) The average temperature of Kodaikanal and Madurai is not same though both are located in Tamil Nadu.

(iv) Study the climatic data given below and answer the questions that follow: [3]

Month	J	F	M	A	M	J	J	A	S	O	N	D
TEMP °C	24.4	25.4	27	30	32	32	31	30.5	30	28	26	25
Rainfall CM	2.8	0.7	0.7	1.5	4.5	5.1	9.5	11.3	12.4	28.1	34.5	30

(a) Calculate the annual rainfall of the given station.

(b) What is the mean temperature of the given station?

(c) Which place does the given data represent? Give reason for your answer.

Question 5

(i) Mr.X, a farmer in the Deccan plateau often struggles with growing water- intensive crops as his land is covered with thick, dark soil. Mr.Y who farms in the eastern coastal plains, enjoys abundant rainfall and has fertile soil. [2]

(a) Identify the types of soil Mr.X and Mr.Y prefer to plough?

(b) Why does Mr.X struggles to grow water intensive crops on his land, despite its ability to retain moisture?

(ii) Give two characteristic features of the soil most suitable for growing tapioca. [2]

- (iii) With reference to alluvial soil answer the following questions: [3]
- How are alluvial soil formed?
 - What type of crops are best suited for cultivation in alluvial soil?
 - Why is deltaic alluvium more fertile than inland alluvium?
- (iv) Give a reason for each of the following: [3]
- In Deccan coastal strips alluvial soil gets darker.
 - Contour bunding is an important method of terrace farming.
 - Leaching exhibits the top soil infertile.

Question 6

- (i) [2]



- Identify the type of vegetation exhibited in the picture.
 - Name an area of India where this kind of vegetation is found.
- (ii) Evaluate the effectiveness of Van-Mahotsav in promoting long term environmental sustainability. [2]
- (iii) Forests are considered the “lungs of the Earth”. Explain why this metaphor is used and describe any two importance of forests. [3]
- (iv) (a) Name any two trees found in the tropical deciduous forest. [3]
- Why is commercial exploitation of tropical evergreen forests challenging?
 - What is the local name for southern mountain forests in the Nilgiris?

Question 7

- (i) Give two reasons to explain why canal irrigation is popular form of irrigation in U.P. [2]
- (ii) Discuss two benefits of using sprinkler irrigation in dry and semi arid regions of India. [2]
- (iii)(a) Name a state which has made it mandatory for every building to harvest water. [3]
- Mention two objectives of rainwater harvesting in India.
- (iv)(a) How is over irrigation responsible for land degradation in Punjab? [3]
- Mention two disadvantages of drip irrigation.

Question 8

- (i) The Indian government has been actively promoting solar energy as a sustainable power source, especially in rural areas where access to conventional electricity is limited. Under the Pradhan Mantri Kisan Urja Suraksha evam Utthaan Mahabhiyan (PM-KUSUM) scheme, farmers are being encouraged to install solar pumps and small solar plants on their fields. This initiative not only aims to reduce dependency on grid electricity but also helps farmers save on diesel costs.

(Edited;Source: <https://pmkusum.mnre.gov.in>) [2]

Mention two ways in which solar energy generation are benefitting.

- (ii) Name two multipurpose projects and name of the rivers on which they have been Constructed in India. [2]
- (iii) Name the mineral ore from which aluminium is extracted. Why is it gaining importance? Name the two leading aluminium producing states in India. [3]
- (iv) Give a geographical reason for each of the following: [3]
 - (a) Many port cities have their own oil refineries.
 - (b) Electrical industry is the major consumer of copper.
 - (c) Biogas is an ideal domestic fuel.

Question 9

- (i) This is the second most important cereal crop. It is the main food crop of North and North -western part of the country. This crop requires cool growing season and a bright sunshine at the time of ripening. [2]
 - (a) Identify the crop.
 - (b) Write two climatic conditions that favour the cultivation of this crop.
- (ii) State one point of difference between commercial farming and subsistence farming. [2]
- (iii) With reference to the cultivation of tea answer the following: [3]
 - (a) Name one important tea growing state of India.
 - (b) How does clonal planting contribute uniformity in crop production?
 - (c) What is the role of shelter belts in the tea plantation?
- (iv) Mr. X visited Burdwan District of West Bengal during Puja holidays. He was mesmerized when he visited the field. He admired the way farmers skilfully maintained the delicate balance of water levels, which nurtured the young crops. [3]
 - (a) Name the crop that Mr.X had seen in the fields.
 - (b) Mention two modern methods by which this above mentioned crop can be sown.

Question 10

- (i) A weaver in a small town of Mysuru is struggling to compete with synthetic fibre industries. [2]
 - (a) Which industry is being talked about?
 - (b) Which state is the leading producer of this industry?
- (ii) Why is there a tendency for the sugar mills to concentrate in southern states of India in recent years? Give two reasons. [2]
- (iii) With reference to Tata Iron and Steel Company answer the following questions: [3]
 - (a) Where is it located?
 - (b) From where does it get its supply of
 - 1. Iron ore
 - 2. Power
- (iv) Give a reason for each of the following: [3]
 - (a) In India the electronic industry contributes a lot to space technology.
 - (b) Petrochemical products are gaining popularity in modern days.
 - (c) Kolkata is an important cotton textile producing centre of West Bengal.

Question 11

- (i) Which is the cheapest means of transport? Mention any one advantage of it other than being cheap. [2]
- (ii) Roadways are always considered more important than any other means of transportation. Give two reasons in support of the statement. [2]
- (iii) (a) Name the agency which is responsible for building and maintaining airports. [3]
 - (b) State any two factors which affect the distribution of railways in the northern Plains.
 - (c) Name any two private sector airlines of India.

- (iv) Give a reason for each of the following: [3]
- (a) Himalayan region has low density network of railways.
 - (b) Airways are quite useful during natural calamities.
 - (c) The peninsular rivers are not ideal for inland water transport.

Question 12

- (i) Mention one way in which waste accumulation has an effect on the following: [2]
- (a) Aquatic life
 - (b) Landscape
- (ii) What do you mean by segregation of waste? How does it ensure safe waste disposal? [2]
- (iii) In a recent initiative, New Delhi's municipal corporation introduced a composting program across residential societies to manage waste. The program encourages residents to separate waste from other household trash and compost it within their premises, reducing the burden on landfills. By processing biodegradable waste locally, the program aims to lower methane emissions and create useful compost for gardens and parks in the community. [3]

(Edited; Times of India, 05.05.2023)

- (a) What type of waste is targeted by the composting program mentioned above?
 - (b) How can composting benefit the environment?
 - (c) State one way in which domestic waste can be made useful.
- (iv)(a) What is open dumping? [3]
- (b) Why are landfills set up far away from the city? (Any two reasons)